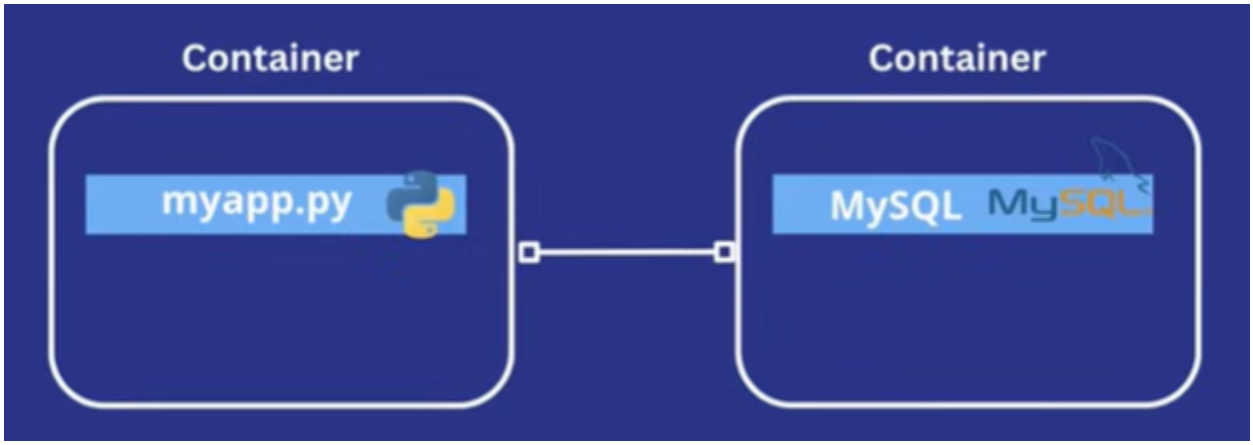
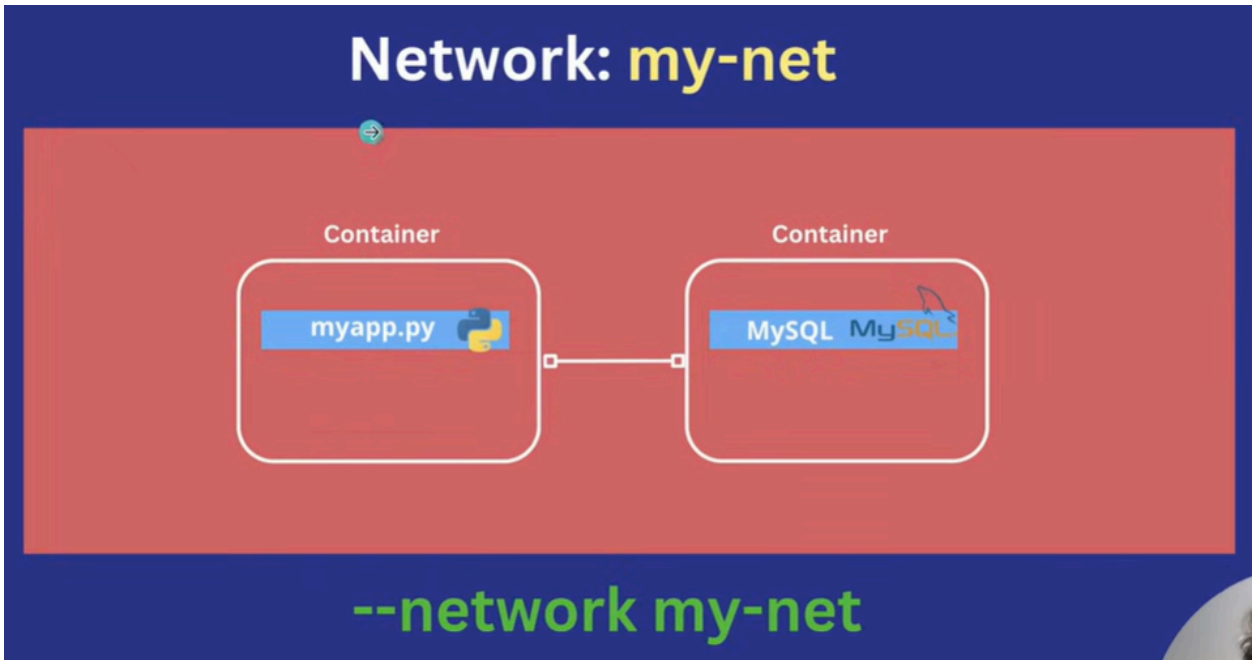
Creating Docker Network



We have used MySQL container IP into the python code. At first, need to build and run mysql always, and then we can create python code image.
It must be like images are created, and only focus on command, so we can directly run it.
Solution is docker network.



Two containers are of the same network part. Support created a docker network named as my-net. Inside the network both the container are running. So they can access each other.
docker ps: shows running containers.

```
\Python Code> docker start mysqldb

\Python Code> docker ps
                CREATED        STATU
ntrypoint.s…"   16 hours ago   Up 1
\Python Code> docker stop mysqldb

\Python Code> docker ps
CREATED   STATUS    PORTS     NAMES
\Python Code>
```

no images are running in container.

To know about docker network:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker network --help
Usage:  docker network COMMAND

Manage networks

Commands:
  connect     Connect a container to a network
  create      Create a network
  disconnect  Disconnect a container from a network
  inspect     Display detailed information on one or more networks
  ls          List networks
  prune       Remove all unused networks
  rm          Remove one or more networks

Run 'docker network COMMAND --help' for more information on a command.
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

To create a network:

```
\Python Code> docker network create my-net
2ad83199f0783de99938a4d0b8fb86
\Python Code>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker network ls
NETWORK ID     NAME        DRIVER    SCOPE
bdebddcb34f3   bridge      bridge    local
2383d8415525   host        host      local
20d15fdc09c3   my-net      bridge    local
3799dd8eb5f5   mynet       bridge    local
6d10c73adbbd   mynetwork   bridge    local
39e07722353c   none        null      local
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

docker run -d --env MYSQL_ROOT_PASSWORD="ghat123@gmail.com" --env MYSQL_DATABASE="employee_db" --name mysqldb_new --network my-net mysql

```
run -d --env MYSQL_ROOT_PASSWORD="ghat123@gmail.com" --env MYSQL_DATA
BASE="employee_db" --name mysqldb_new --network my-net mysql
12d08a9550bb84de88c8755a23e8f87fc7a40da0e9f851628c119035a6c4ace6
```

```
de> docker ps
      CREATED          STATUS          PORTS                    NAMES
s…"   4 minutes ago    Up 5 minutes    3306/tcp, 33060/tcp      mysqldb_new
de>
```

Changes in python code:

```
file    PyCode.py
de.py
1  import pymysql
2
3  # ------------------ DB CONFIG ------------------
4  DB_HOST = "mysqldb_new"
5  DB_USER = "root"
```

```
S   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   PLAYWRIGHT

 --env MYSQL_ROOT_PASSWORD="ghat123@gmail.com" --env MYSQL_DATABASE="employee_db" --name mysqldb_new --networ
9550bb84de88c8755a23e8f87fc7a40da0e9f851628c119035a6c4ace6
Users\kiran\OneDrive\Desktop\Python Code> docker ps
NER ID   IMAGE     COMMAND                  CREATED        STATUS         PORTS                    NAMES
9550bb   mysql     "docker-entrypoint.s…"   4 minutes ago  Up 5 minutes   3306/tcp, 33060/tcp      mysqldb_new
Users\kiran\OneDrive\Desktop\Python Code>
```

Check images before build:

```
ktop\Python Code> docker images -a

  ID             DISK USAGE   CONTENT SIZE   EXTRA
  24e450bbd24f       1.27GB         283MB     U
  b90c1a14ca28       2.11GB         491MB
  b90c1a14ca28       2.11GB         491MB
  2a5af4bc4bb9       1.66GB         429MB
  2c4ce40d1bb1       1.63GB         418MB
  f9cc710aa4ab       1.63GB         418MB
ktop\Python Code> docker build .
SHED
nition from Dockerfile
```

After build:

```
ktop\Python Code> docker images -a

  ID             DISK USAGE   CONT
  24e450bbd24f       1.27GB
  b90c1a14ca28       2.11GB
  b90c1a14ca28       2.11GB
  42f2034df309       1.66GB
  2a5af4bc4bb9       1.66GB
  2c4ce40d1bb1       1.63GB
  f9cc710aa4ab       1.63GB
ktop\Python Code>
```

Run the docker image under network:

```
hon Code> docker run -it --rm --network my-net 42f2034df309
```

```
✅ Database connected successfully!


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 3

--- View Employees ---
Show all? (y/n): y
⚠️No employees found.


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 1

--- Add Employee ---
Enter Name (required): name1 name11
Enter Phone (required): 606780
Enter Email (optional): dfnadf@we.co
Enter Salary (optional): 2343
Enter Aptitude Score (0-10) (required): 5.423
✅ Employee added successfully!
```

```
===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 1

--- Add Employee ---
Enter Name (required): name2 name12
Enter Phone (required): 4523
Enter Email (optional): 24234@24323.com
Enter Salary (optional): 452342
Enter Aptitude Score (0-10) (required): 7.23342
✅ Employee added successfully!


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 3

--- View Employees ---
Show all? (y/n): y

--- Employee Records ---
(1, 'name1 name11', 'dfnadf@we.co', '606780', 2343.0, 5.423)
(2, 'name2 name12', '24234@24323.com', '4523', 452342.0, 7.23342)
```

When we are dealing with multiple containers, you can assign these may containers to common network.

In python code,

```
erfile        PyCode.py
ode.py
1  import pymysql
2
3  # ------------------- DB CON
4  DB_HOST = "mysqldb_new"
5  DB_USER = "root"
```

Simply provide the host, no need to change in file, or creating build. Only thing happens in command line.